

Lukas Helmig

PhD Student in Theoretical Physics

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📍 Osnabrück, Germany

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Education

PhD	Osnabrück University , Physics • Investigating equilibration in disordered systems.	Osnabrück, Germany 2025 – 2028
MS	University of Cologne , Physics • Specialized in solid state theory and quantum information. • Thesis on "Restricted Boltzmann Machines and Hopfield Networks". • Top of the year.	Cologne, Germany 2022 – 2025
	Ecole Normale Supérieure de Paris , Physics • Participated in the Erasmus exchange program, focusing on advanced topics in theoretical physics.	Paris, France 2023 – 2024
BS	Osnabrück University , Physics • Thesis on "Influence of interactions on the efficiency at maximal power of a microscopical thermodynamic machine". • Top of the year. Finished with honors.	Osnabrück, Germany 2018 – 2021

Experience

Faculty of Physics, Osnabrück University , Research Assistant Worked on the VRnano project, a physical development environment for better control of laboratory experiments.	Osnabrück, Germany 2021 – 2023 2 years
Faculty of Mathematics, Osnabrück University , Teaching Assistant Tutorial lectures and exam grading.	Osnabrück, Germany 2019 – 2021 2 years

Awards

Honors branch Awarded for outstanding academic achievement University of Cologne	2022
Niedersachsen Scholarship Awarded for outstanding academic achievement Niedersachsen Ministry of Science and Culture	2021
Deutschlandstipendium Awarded for outstanding academic achievement from 2019 to 2021 Osnabrück University	2019

Teaching

Pre-course in Mathematics , Lecturer • Taught a 2-week pre-course in mathematics for first-year physics students. • Covered topics such as linear algebra, calculus, and probability theory.	Osnabrück, Germany 2026
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Theoretical Physics III, Teaching Assistant

- Advanced topics in classical mechanics and electrodynamics.
- Held weekly exercise classes and provided feedback to students.

Osnabrück, Germany
2026

Theoretical Physics IV, Teaching Assistant

- Advanced topics in quantum mechanics and statistical mechanics.
- Held weekly exercise classes and provided feedback to students.
- The course placed in the top 14% of all courses at Osnabrück University in the student evaluation.

Osnabrück, Germany
2025 – 2026

Outreach

DOPPLERS, Volunteer

DOPPLERS is a yearly theoretical physics competition held at different german universities and supported by the German Physical Society (DPG).

- Organized the academic program including two speakers and held a workshop on decision-making myself.
- Sought and acquired sponsoring for the catering.
- Volunteered and helped with the event in general.

Osnabrück, Germany
2026

Deutsche Schülerakademie, Course Instructor

The course was held in Jülich, Germany, and was part of the Deutsche Schülerakademie program.

- Developed a 2-week course (50 hours) on quantum computing for high school students.
- Conducted lectures and exercise lessons
- Guided students through a final project phase with an accompanying poster session.

Jülich, Germany
2025